PYL122 Assignment 9
Tobias Toll

Instructions: *Solve the problems during the tutorial session. Hand in the solutions to the TA one week after the last tutorial. Submit hardcopy to MS207c-19, or soft copy to gaurab.Kumar.Dash@physics.iitd.ac.in. Assignment will be considered for marks if handed in on time.*

1. **Addition of Angular Momenta**
Let $\hat{\vec{J}} = \hat{\vec{J}}_1 + \hat{\vec{J}}_2$ where $\hat{\vec{J}}_1$ and $\hat{\vec{J}}_2$ are angular momentum operators.
(a) Show that this implies that $\hat{\vec{J}}$ is an angular momentum operator.
(b) Show that $\hat{J}_1^2$, $\hat{J}_2^2$, $\hat{J}^2$, and $\hat{J}_z$ have common eigenstates.
(c) Show that these are not eigenstates of $\hat{J}_{1z}$ and $\hat{J}_{2z}$.
(d) We may write the combined angular momentum eigenstates of $\hat{J}_1^2, \hat{J}_{1z}$ and $\hat{J}_2^2, \hat{J}_{2z}$as $|j_1 m_1; j_2 m_2\rangle \equiv |j_1, m_1\rangle |j_2, m_2\rangle$. We write the eigenstates to $\hat{J}_1^2$, $\hat{J}_2^2$, $\hat{J}^2$, and $\hat{J}_z$ as $|j_1 j_2; jm\rangle$
Show that $m = m_1 + m_2$
(e) Show that $|j_1 - j_2| \leq j \leq j_1 + j_2$
(f) We can always express the eigenstates in one basis as a linear combination of the eigenstates in a different basis. Here $|j_1 j_2; jm\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} \delta_m^{m_1+m_2} \alpha(j_1, m_1, j_2, m_2) |j_1 m_1; j_2 m_2\rangle$
With $\alpha(j_1, m_1, j_2, m_2) = \langle j_1 m_1; j_2 m_2 | j_1 j_2; jm\rangle$
Find the combined eigenfunctions for two spin 1/2 particles $|\frac{1}{2}\frac{1}{2}; sm\rangle$ expressed is the original spin bases $|\frac{1}{2}, m_1\rangle$ and $|\frac{1}{2}, m_2\rangle$ (for $j = \frac{1}{2}$ we usually call it spin and denote it by s).
(g) Calculate the Clebsch-Gordan coefficients for combining the spins of two spin 1/2 particles.
***(h)** Calculate the Clebsch-Gordan coefficients for combining an electron's spin with its angular momentum with $l = 1$. Proceed similarly as with the case of two spin 1/2 particles:
1. Find all possible j-values. Take the state with largest j value and largest m-value. This must correspond to the state with largest m_l and m_s values in the uncoupled base. One coefficient has been found. By symmetry, the state with smallest m-value has also been found.
2. Act on the coupled state in 1. with J_- and the uncoupled state with $L_- + S_-$ to find the next state. By symmetry you have now found the CG-coefficients for two more states.
3. The remaining states are found by applying the orthonormality condition.

2. **Some fundamentals**
You have previously found that the eigenvalues to the momentum operator (in one dimension) is $|p\rangle = \psi_p(x) = e^{ipx/\hbar}$.
(a) Convince yourself that the eigenvalues to the position operator $\hat{x}$ for measuring the position of a particle at $x = x_0$ is given by $|x\rangle = \phi_x(x) = \delta(x - x_0)$ where $\delta(x - x_0)$ is the Dirac delta-function with the property $\int_{-\infty}^{\infty} \mathrm{d}x \delta(x - x_0) f(x) = f(x_0)$
As you may notice, neither $\phi_x(x)$ nor $\psi_p(x)$ are normalisable. This is related to the fact that the free particle solutions are also non-normalisable.

(b) In the course we have been expressing wavefunctions in coordinate space $\psi = \psi(x)$. We may also express them in momentum space via a Fourier transform: $\tilde{\psi}(k) = \int \mathrm{d}x e^{-2\pi i k x}\psi(x)$ where the momentum is as usual $p = hk$.
What is the overlap between the momentum eigenvalues and $\psi(x)$: $\langle p | \psi \rangle$?

(c) Let our system be in a state $\psi(x) = \dfrac{1}{\sigma_x\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma_x}}$, a Gaussian with width $\Delta x = \sqrt{\sigma_x}$.

Find $\tilde{\psi}(p)$ and Δp. What is $\Delta x \Delta p$? What happens to the two states as $\sigma_x \to 0$?

You may read more about this in Shankar, Chapter 1.

3. **The Hydrogen Atom**

We have found the solutions to the electron wave function in the hydrogen atom as

$$\Psi(r,\theta,\phi) = R(r)\Upsilon_l^{m_l}(\theta,\phi), \text{ with } R(r) = \frac{e^{-\kappa r}}{r}\sum_{q=l+1}^{n} b_q r^q \text{ with } b_{q+1} = \frac{2q\kappa - 2/a}{q(q+1) - l(l+1)} b_q$$

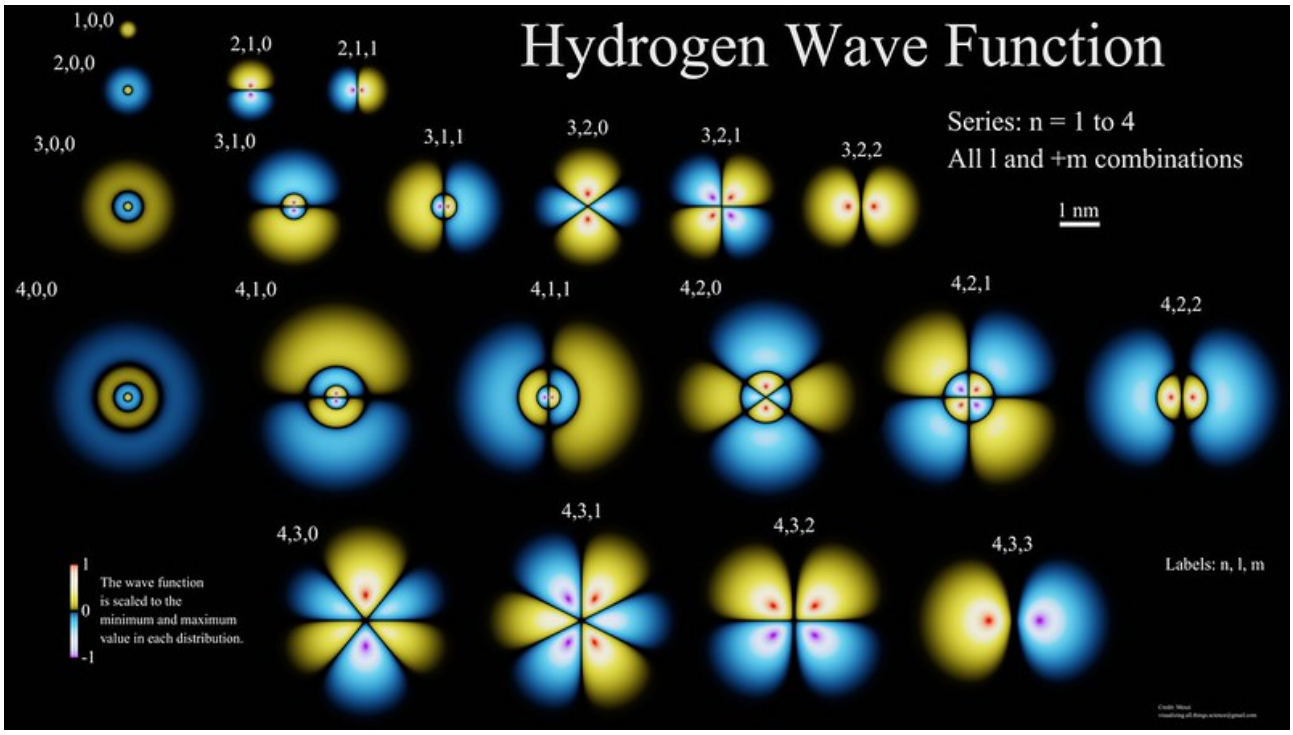
Here, $n = 1,2,3\ldots$ is the *principal quantum number* (a better name would be the *radius quantum number*), $l < n$, and from normalisation we found that $b_1 = \sqrt{\dfrac{4}{a^3}}$. Here, $\kappa^2 = -\dfrac{2mE_n}{\hbar^2} > 0$, and the Bohr radius $a = \dfrac{4\pi\epsilon_0\hbar^2}{me^2}$, with the reduced mass $m = \dfrac{m_e m_p}{m_e + m_p}$.

(a) Write down the complete 3D wave-function for the ground state, as well as its energy and draw $R(r)$ as a function of r

(b) Find the expectation value of r in the ground state.

(c) Find the most common value of r in the ground state.

(d) Find the wave functions of the first excited states and its energy.

(e) How many possible states are there for the electron in the ground state and in the first excited state? Can you relate this to the periodic table of elements?

Hint: The probability of finding the electron in the radius interval between r and $r + \mathrm{d}r$ is given by $\mathrm{d}P = |R(r)|^2 r^2 \mathrm{d}r$